AS-2011	AS-2020
01 - list	01/02(CMLDS) - list
res - 2	1 - 2
reserved	0x00
octet 1 bit 0-5	new twoStepFlag o1b1
reserved	All 0's
0x05	0x00

PTP Messages 10.6

4-Byte align Byte aligned

Header 10.6.2 — 34 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
gPTP Message Header								34	
majorSdoId				messageType				1	0
minorVersionPTP				versionPTP				1	1
messageLength								2	2
domainNumber								1	4
minorSdoId								1	5
flags								2	6
correctionField								8	8
messageTypeSpecific								4	16
sourcePortIdentity								10	20
sequenceId								2	30
constrolField								1	32
logMessageInterval								1	33

Announce 10.6.3 — 68 + 8*N Bytes

7	6	5	4	3	2	1	0	Octets	Offset
Announce Message: 68 + N*8								68 + 8 * (stepsRemoved+1)	
header (see 10.6.2)								34	0
reserved								10	34
currentUtcOffset								2	44
reserved								1	46
grandmasterPriority1								1	47
grandmasterClockQuality								4	48
grandmasterPriority2								1	52
grandmasterIdentity								8	53
stepsRemoved								2	61
timeSource								1	63
path trace TLV: tlvType 0x08								2	64
path trace TLV: lengthField								2	66
path trace TLV: pathSequence								N * 8	68

Signaling Message changes with 802.1ASdm: Can be more than on TLV and may also potentially contain other TLVs

Singaling 10.6.4 — 60 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
Signaling Message								60	
header (see 10.6.2)								34	0
targetPortIdentity = all ones								10	34
one of: message interval request TLV, gPTP-capable TLV, or gPTP-capable message interval request TLV								16	44

11.4.3 — 76 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
1-Step Sync								76	
header (see 11.4.2)								34	0
originTimestamp								10	34
Follow_Up information TLV: tlvType								2	44
Follow_Up information TLV: lengthField								2	46
organizationId								3	48
organizationSubType								3	51
cumulativeScaledRateOffset								4	54
gmTimeBaseIndicator								2	58
lastGmPhaseChange								12	60
scaledLastGmFreqChange								4	72

11.4.3 — 44 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
2-Step Sync								44	
header (see 11.4.2)								34	0
reserved								10	34

11.4.4.1-3 — 76 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
Follow-Up								76	
header (see 11.4.2)								34	0
preciseOriginTimestamp								10	34
Follow_Up information TLV: tlvType = 0x03								2	44
lengthField = 28								2	46
organizationId = 00-80-C2								3	48
organizationSubType = 1								3	51
cumulativeScaledRateOffset								4	54
gmTimeBaseIndicator								2	58
lastGmPhaseChange								12	60
scaledLastGmFreqChange								4	72

11.4.4.4 — 32 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
Drift_Tracking TLV: tlvType = 0x03								2	76
lengthField = 32								2	78
organizationId = 00-80-C2								3	80
organizationSubType = 6								3	83
syncEgressTimestamp								12	86
syncGrandmasterIdentity								8	98
syncStepsRemoved								2	106
rateRatioDrift								4	108

11.4.5 — 54 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
Pdelay_Req Message								54	
header (see 11.4.2)								34	0
reserved								10	34
reserved								10	44

11.4.6 — 54 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
Pdelay_Resp Message								54	
header (see 11.4.2)								34	0
requestReceiptTimestamp								10	34
requestingPortIdentity								10	44

11.4.7 — 54 Bytes

7	6	5	4	3	2	1	0	Octets	Offset
Pdelay_Resp_Follow_Up Message								54	
header (see 11.4.2)								34	0
responseOriginTimestamp								10	34
requestingPortIdentity								10	44

Msg type	Msg class	Value
Sync	Event	0x0
Pdelay_Req	Event	0x2
Pdelay_Resp	Event	0x3
Follow_Up	General	0x8
Pdelay_Resp_Follow_Up	General	0xA
Announce	General	0xB
Signaling	General	0xC

Table 10-9—Values of flag bits

Octet	Bit	Message types	Name	Value
0	0	All	alternateMasterFlag in Announce, Sync, Follow_Up, and Delay_Resp messages	Not used in this standard; transmitted as FALSE and ignored on reception
0	1	Sync, Pdelay_Resp	twoStepFlag	*For Sync messages:* a) For a one-step transmitting PTP Port (see 11.1.3 and 11.2.13.9), the value is FALSE. b) For a two-step transmitting PTP Port, the value is TRUE. *For Pdelay_Resp messages:* The value is transmitted as TRUE and ignored on reception.
0	2	All	unicastFlag	Not used in this standard; transmitted as FALSE and ignored on reception
0	3	All	Reserved	Not used by IEEE Std 1588–2019; reserved as FALSE and ignored on reception
0	4	All	Reserved	Not used by IEEE Std 1588–2019; reserved as FALSE and ignored on reception
0	5	All	PTP profileSpecific 1	Not used in this standard; transmitted as FALSE and ignored on reception
0	6	All	PTP profileSpecific 2	Not used in this standard; transmitted as FALSE and ignored on reception
0	7	All	Reserved	Not used in this standard; transmitted as FALSE and ignored on reception
1	0	Announce	leap61	The value of the global variable leap61 (see 10.3.9.4)
1	1	Announce	leap59	The value of the global variable leap59 (see 10.3.9.5)
1	2	Announce	currentUtcOffsetValid	The value of the global variable currentUtcOffsetValid (see 10.3.9.6)
1	3	Announce	ptpTimescale	The value of the global variable ptpTimescale (see 10.3.9.7)

Table 10-9—Values of flag bits *(continued)*

Octet	Bit	Message types	Name	Value
1	4	Announce	timeTraceable	The value of the global variable timeTraceable (see 10.3.9.8)
1	5	Announce	frequencyTraceable	The value of the global variable frequencyTraceable (see 10.3.9.9)
1	6	All	Reserved	Not used by IEEE Std 1588–2019; reserved as FALSE and ignored on reception
1	7	All	Reserved	Not used in this standard; reserved as FALSE and ignored on reception

IEEE 802.1AS uses the same multicast MAC address for all messages.
This can be done because in an 802.1AS newtork, each device must support IEEE 802.1AS.

General messages are not required to be timestamped.

11.3.3 VLAN tag

A frame that carries an IEEE 802.1AS message shall not have a VLAN tag nor a priority tag (see 3.184 of IEEE Std 802.1Q-2018).

11.3.4 Addresses

The destination address of Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be the reserved multicast address given in Table 11-3.

Table 11-3—Destination address for Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages

Destination address
01-80-C2-00-00-0E
NOTE—This address is taken from Table 8-1, Table 8-2, and Table 8-3 of IEEE Std 802.1Q-2018.

All Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall use the MAC address of the respective egress physical port as the source address.

11.3.5 EtherType

The EtherType of Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be the EtherType given in Table 11-4.

Table 11-4—EtherType for Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages

EtherType
88-F7

The destination address of the Announce and Signaling messages shall be the reserved multicast address given in Table 10-4 unless otherwise specified in a media-dependent clause (see 12.2 and 16.2).

Table 10-4—Destination address for Announce and Signaling messages

Destination address
01-80-C2-00-00-0E
NOTE—This address is taken from Table 8-1, Table 8-2, and Table 8-3 of IEEE Std 802.1Q-2018.

NOTE—Frames whose destination address is the address of Table 10-4 are never forwarded, according to IEEE 802.1Q protocol. Use of this address is shared by IEEE 802.1AS and other IEEE 802.1 protocols.

10.5.4 EtherType

The EtherType of the Announce and Signaling messages shall be the EtherType given in Table 10-5.

Table 10-5—EtherType for Announce and Signaling messages

EtherType
88-F7

NOTE—This EtherType is used for all PTP messages.

IEEE 1588 uses 2 MAC addresses:

All except Pdelay messages	01-1B-19-00-00-00
Peer delay messages	01-80-C2-00-00-0E

The first MAC address is a "normal2 multicast address that will flood the network.
The second addres is from a reserved area thata is not forwarded by switches over the network, only to the lo

ɔcal Switch CPU.